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Discrete Linear Canonical Transform of Finite Chirps

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Abstract

Linear canonical transformation (LCT) as an Fourier transform and fractional fourier transform's generalization, is an effective tool for non-stationary signals and has more flexibility, chirp signal can be looked as a typical non-stationary signal. In this letter, we derived closed-form expressions for the DLCT of a finite chirp. It is shown that the DLCT of a finite chirp is zeros at some points, and the number of zeros is related to chirp rate $\bar{\alpha}$, the parameter of DLCT $\tilde{\alpha}$, and the root of unity N . And if we choose proper DLCT parameters, the finite chirp is again a finite chirp or the original chirp except for a coefficient.

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Discrete linear canonical transform (DLCT); Finite chirp; Gauss sum; Jacobi symbol.

1. Introduction

Fourier transform is proved to be the ideal tool for processing stationary signals, but in the non-stationary signal processing of great limitations. To solve this problem, people propose and develop a series of new signal analysis theories: short-time Fourier transform, time-frequency transform, Gabor transform, wavelet transform, Radon-Wigner transform, fractional Fourier transform, the linear canonical transformation (LCT) and so on [1][2]. Among them, the LCT is considered as one of the most effective non-stationary signal analysis and processing tools, with more flexibility.

Chirp signal is a typical non-stationary signal, which widely appears in systems such as communication, radar, sonar, biomedicine and earthquake exploring system and also be used as signal model for a lot of natural phenomena, therefore, properties of chirp signal became the points of focus of people. For now, people has made a series of important achievements for continuous chirp signal analysis, parameter detection and estimation [3][4], while few for the discrete chirp signals, and the existing research is also

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based on classic Fourier transform [5]. Therefore, it's necessary and significance to analysis the properties of the finite chirps in the transform domain.

Based on above consideration, this letter focuses on an important class of discrete chirps-finite chirps associated with the LCT, we firstly study the properties of finite chirps in discrete linear canonical transformation (DLCT), and then analyze the nature of finite chirps in DLCT and discuss the applications of the achieved results.

The remainder of this paper is as follows. In Section 2, we introduce finite chirps and the DLCT. In Section 3, we present the DLCT of a finite chirp and discuss the applications of the achieved results. In Section 4, we give some conclusions.

2. Preliminaries

2.1. Finite chirps

In [5], the finite chirp and its discrete Fourier transform (DFT) was proposed. Set $e_N(n) := e^{2i\pi(n/N)}$ the N th root of unity. The discrete chirp is defined as

$$x(n) = e_N(\frac{1}{2}\bar{a}n^2 + bn), \quad 0 \leq n < N. \quad (1)$$

Where $\bar{a} = a/N$ is the normalized version of the discrete chirp rate, a is the discrete chirp rate, and b is the discrete carrier frequency, $a, b \in R$. Consider a discrete chirp shifted by N

$$x(n+N) = x(n)e(\frac{1}{2}a+b)e_N(an), \quad 0 \leq n < N. \quad (2)$$

It follows that the discrete chirp is periodic modulo N , i.e., $x(n+N) = x(n)$, iff

$$\bar{a} \in Z \text{ and } \frac{a}{2} + b \in Z.$$

A discrete chirp satisfying periodic modulo N is called the finite chirp.

Then, the DFT of a finite chirp $x(n)$ can be expressed by the formula [5]

$$X_N(k) = \sum_{n \bmod N} e_N(\frac{\alpha}{2}n^2 + (b+k)n) = \sum_{n \bmod N} e_N(\alpha n^2 + \beta n) \quad (3)$$

and, $\alpha = \bar{a}/2$, $\beta = b+k$.

Until now, people has studied the DFT of finite chirps and got a good result [5], while there's no research to it's DLCT. So, we will study the DLCT of finite chirp signals in section 4.

2.2. DLCT

The three parameter LCT of a signal $f(x)$ is denoted as [8][9]

$$L_{\alpha, \beta, \gamma}\{f(x)\}(x') = \int_{-\infty}^{+\infty} f(x) e^{(i\pi/2)(\alpha x^2 - 2\beta xx' + \gamma x'^2)} dx,$$

Here, $f(x)$ is a function in a domain x and the output is in a domain x' , e.g. time and frequency in the case of the FT, and α , β and γ are the parameters of the transform.

In DSP the LCT is an important tool, and it can be calculated analytically for many functions. A discrete version of the LCT will be used in the real applications, so we require a discrete version for digital computation. To this problem, many researches have proposed several DLCTS and they are the equal forms [8][10][11][12]. In order to get the same output range as input range for function $f(x)$, we use scale-preserving DLCT proposed in [10] in this letter.

Scale-preserving DLCT is defined as in [10]:

$$L_{\alpha, \beta, \gamma}^{N, T}\{f(nT)\}(mT_y) = \sum_{n=-N/2}^{n=N/2} f(nT) W_T^{n, m} \quad (4)$$

where the kernel is

$$W_T^{n, m} = e^{i\pi T^2/2(\alpha n^2 - 2\beta nm + \gamma m^2)} = e^{2\pi i(\frac{h^2\alpha}{4} \times \frac{n^2}{N} - \frac{h^2\beta}{2} \times \frac{nm}{N} + \frac{h^2\gamma}{4} \times \frac{m^2}{N})},$$

and h is $\sqrt{NT}$.

The input range and output range are $-NT/2$ to $NT/2$ in this DLCT.

Set

$$\bar{\alpha} = \frac{h^2\alpha}{4} = \frac{NT^2\alpha}{4}, \bar{\beta} = -\frac{h^2\beta}{2} = -\frac{NT^2\beta}{2}, \gamma = \frac{h^2\gamma}{4} = \frac{NT^2\gamma}{4},$$

and choose $\bar{\alpha}, \bar{\beta}, \bar{\gamma} \in Z$. Then the kernel of DLCT is changed into

$$W_T^{n,m} = e^{\frac{2\pi i(\bar{\alpha}x^2}{N} - \bar{\beta}x\frac{nm}{N} + \bar{\gamma}x\frac{m^2}{N})} = e_N(\bar{\alpha}n^2 + \bar{\beta}nm + \bar{\gamma}m^2).$$

and

$$\alpha = \frac{4}{NT^2}, \beta = \frac{-2}{NT^2}, \gamma = \frac{4}{NT^2}.$$

3. The DLCT of finite chirps

3.1. The DLCT of finite chirps

Take x to be a discrete chirp

$$x(n) = e_N(\frac{1}{2}\bar{a}n^2 + bn), \quad 0 \leq n < N.$$

satisfying periodic modulo N .

According to the definition of DLCT above, then DLCT of a finite chirp $x(n)$ is

$$\begin{aligned} L_{\alpha,\beta,\gamma}^{N,T}\{x(n)\}(m) &= \sum_{n \bmod N} x(n)W_T^{n,m} = \sum_{n \bmod N} e_N(\frac{1}{2}\bar{a}n^2 + bn)e_N(\bar{\alpha}n^2 + \bar{\beta}nm + \bar{\gamma}m^2) \\ &= e_N(\bar{\gamma}m^2) \sum_{n \bmod N} e_N[(\frac{1}{2}\bar{a} + \bar{\alpha})n^2 + (b + \bar{\beta}m)n]. \quad (0 \leq m < N) \end{aligned} \quad (5)$$

Set $\tilde{\alpha} = \frac{1}{2}\bar{a} + \bar{\alpha}$, $\tilde{\beta} = b + \bar{\beta}m$,

Then

$$L_{\alpha,\beta,\gamma}^{N,T}\{x(n)\}(m) = e_N(\bar{\gamma}m^2) \sum_{n \bmod N} e_N(\tilde{\alpha}n^2 + \tilde{\beta}n) =: S_1 S_2, \quad (6)$$

Where $S_1 = e_N(\bar{\gamma}m^2)$ is a constant for given m , and

$$S_2 = \sum_{n \bmod N} e_N(\tilde{\alpha}n^2 + \tilde{\beta}n). \quad (7)$$

Because of $\bar{a} \in Z$ and $\bar{\alpha} \in Z$, so $\tilde{\alpha} \in Z$ or $(Z + \frac{1}{2})$. For given m , S_2 can be seen as a form of DFT of a finite chirp. Therefore, according to classical results in [5], the DLCT of a finite chirp $x(n)$ is given by the following theorem.

Theorem 1: Take $x(n)$ to be a finite chirp, and set the parameters as above, then DLCT of $x(n)$ has two cases, $\tilde{\alpha} \in Z$ and $\tilde{\alpha} \in (Z + \frac{1}{2})$. For $\tilde{\alpha} \in Z$, $(\tilde{\alpha}, N) = c$, $1 \leq c \leq \tilde{\alpha}$. For $\tilde{\alpha} \in (Z + \frac{1}{2})$, $(2\tilde{\alpha}, 2N) = 2c = c'$, $1 \leq c' \leq 2\tilde{\alpha}$.

Set $P = N/c$, $\mathcal{G} = \tilde{\alpha}/c$, $f = \tilde{\beta}/c$, and $1 \leq \mathcal{G}_1 < P$ such that $\mathcal{G}_1 \equiv 1 \pmod{P}$. Then

1) For $\tilde{\alpha} \in Z$, the DLCT of $x(n)$ is given by the formula

$$\frac{L_{\alpha,\beta,\gamma}^{N,T}\{x(n)\}(m)}{e_N(\bar{\gamma}m^2)e_{4P}[-\mathcal{G}(\mathcal{G}_1 f)^2]} = \begin{cases} c\mathcal{G}(\mathcal{G}, P) & \mathcal{G}_1 f \text{ even} \\ \frac{c}{2}[G(\mathcal{G}, 4P) - 2G(\mathcal{G}, P)] & \mathcal{G}_1 f \text{ odd} \\ 0 & f \notin Z \end{cases} \quad (8)$$

where $G(m, n)$ is the quadratic Gauss sum [5], and

2) For $\tilde{\alpha} \in (Z + \frac{1}{2})$, the DLCT of $x(n)$ is given by the formula

$$\frac{L_{\alpha,\beta,\gamma}^{N,T}\{x(n)\}(m)}{e_N(\bar{\gamma}m^2)e_{4P}[-\mathcal{G}(\mathcal{G}_1 f)^2]} = \begin{cases} \left(\frac{P}{\mathcal{G}}\right)c\sqrt{P}(1 + i^{\mathcal{G}}) & f \in Z \\ 0 & \text{else} \end{cases} \quad (9)$$

where $\left(\frac{p}{\gamma}\right)$ is the Jacobi symbol [7].

In the remainder of this letter, we discuss the properties of DLCT of a finite chirp and give some exceptions.

3.2. The applications

(1) $\tilde{\alpha} \in \mathbb{Z}$

First, consider zeros of $L_{\alpha, \beta, \gamma}^{N, T} \{x(n)\}(m)$. Since $f \in \mathbb{Z}$ for N values of m , $L_{\alpha, \beta, \gamma}^{N, T} \{x(n)\}(m)$ has at least $(c-1)N/c$ zeros.

Next, consider a special case, take $\bar{\alpha} = 1 - \frac{1}{2}\bar{a}$, and $c=1$. When the condition $\bar{\alpha} = 1 - \frac{1}{2}\bar{a}$ is met, $\bar{\alpha} = 1$. Then

$$S_2 = \sum_{n \bmod N} e_N(n^2 + \tilde{\beta}n) = \sum_{n \bmod N} e_N[(n + \frac{\tilde{\beta}}{2})^2 - \frac{\tilde{\beta}^2}{4}] = e_{4N}(-\tilde{\beta}^2) \sum_{n \bmod N} e_N\left(n + \frac{\tilde{\beta}}{2}\right)^2. \quad (10)$$

Given

$$S_3 = \sum_{n \bmod N} e_N\left(n + \frac{\tilde{\beta}}{2}\right)^2.$$

For an even $\tilde{\beta}$,

$$S_3 = \sum_{n \bmod N} e_N\left(n + \frac{\tilde{\beta}}{2}\right)^2 = \sum_{n \bmod N} e_N(n^2) = G(1, N).$$

For an odd $\tilde{\beta}$,

$$\begin{aligned} S_3 &= \sum_{n \bmod N} e_N\left(n + \frac{\tilde{\beta}}{2}\right)^2 = \sum_{n \bmod N} e_N\left(n + \frac{1}{2}\right)^2 \\ &= \frac{1}{2} \sum_{n \bmod 4N} e_{4N}(n^2) = \frac{1}{2} \left\{ \sum_{n \bmod 4N} e_{4N}(n^2) - \sum_{n \bmod 4N} e_{4N}(n^2) \right\} \\ &= \frac{1}{2} \left\{ \sum_{n \bmod 4N} e_{4N}(n^2) - 2 \sum_{n \bmod N} e_N(n^2) \right\} = -G(1, N). \end{aligned}$$

According to the above derivation leads to the following formula

$$\frac{L_{\alpha, \beta, \gamma}^{N, T} \{x(n)\}(m)}{e_N(\tilde{\gamma}m^2)e_{4P}(-\tilde{\beta}^2)} = (-1)^{\tilde{\beta}} G(1, N), \quad (11)$$

and

$$G(1, N) = \begin{cases} 0, & N \equiv 0 \pmod{4} \text{ or } N \equiv 2 \pmod{4} \\ \sqrt{N}, & N \equiv 1 \pmod{4} \\ i\sqrt{N}, & N \equiv 3 \pmod{4} \end{cases}$$

It follows that when $\bar{\alpha} = 1 - \frac{1}{2}\bar{a}$ and N is odd, then the DLCT of a finite chirp is a finite chirp with an appropriate scaled chirp rate and carrier frequency; when N is even, then the DLCT of a finite chirp is zeros. When $\bar{\alpha} = 1 - \frac{1}{2}\bar{a}$, $\tilde{\beta} = -2$, $\tilde{\gamma} = \frac{1}{2}\bar{a} - 1$ and N is odd, then the DLCT of a finite chirp is the original finite chirp except for a coefficient.

$$L_{\alpha, \beta, \gamma}^{N, T} \{x(n)\}(m) = G(1, N) e_{4N}(-b^2) e_N\left(\frac{1}{2}\bar{a}n^2 + bn\right) \quad (12)$$

(2) $\tilde{\alpha} \in (\mathbb{Z} + \frac{1}{2})$

First, consider zeros of $L_{\alpha, \beta, \gamma}^{N, T} \{x(n)\}(m)$. Since $f \in \mathbb{Z}$ for N values of m , $L_{\alpha, \beta, \gamma}^{N, T} \{x(n)\}(m)$ has at least $(c - \frac{1}{2})N/c$ zeros.

Next, consider a special case, take $\bar{\beta} = -2\tilde{\alpha}$. When the condition $\bar{\beta} = -2\tilde{\alpha}$. Then

$$\frac{L_{\alpha, \beta, \gamma}^{N, T} \{x(n)\}(m)}{e_N[(\tilde{\gamma} - \tilde{\alpha})m^2]e_N(bm)e_{4N}(-\frac{b^2}{\tilde{\alpha}})} = \begin{cases} \left(\frac{p}{g}\right)c\sqrt{P}(1+i^g) & f \in \mathbb{Z} \\ 0 & \text{else} \end{cases} \quad (13)$$

is a finite chirp with an appropriately scaled chirp rate and carrier frequency except for zeros at some points. When $\bar{\beta} = -2\tilde{\alpha}$ and $\bar{\gamma} = \frac{1}{2}\bar{a} + \tilde{\alpha}$, then the DLCT of a finite chirp is the original finite chirp except for a coefficient and zeros at some points.

$$\frac{L_{\alpha,\beta,\gamma}^{N,T}\{x(n)\}(m)}{e_N(\frac{1}{2}\bar{a}n^2 + bn)e_{4N}(-\frac{b^2}{\bar{a}})} = \begin{cases} (\frac{p}{g})c\sqrt{P}(1+i^g) & f \in Z \\ 0 & \text{else} \end{cases} \quad (14)$$

4. Conclusion

Closed-form expressions for DLCT of a finite chirp have been derived. It was shown that when the condition $\tilde{\alpha} \in Z$ is met, the DLCT of a finite chirp has at least $(c-1)N/c$ zeros; when the condition $\tilde{\alpha} \in (Z + \frac{1}{2})$ is met, the DLCT of a finite chirp has at least $(c - \frac{1}{2})N/c$ zeros. If we choose proper DLCT parameters, the finite chirp is still a finite chirp or original finite chirp except for a coefficient.

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